A data factory (DataOps) for statistical classification task level, data and application logic modeling

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AUTOMATION COST AND STATISTICS

For each use case compute the following**.**

Normalized salaries

See the European ICT Professional role profiles and decide the job role involved in each use case, matching the related skills to the use case steps.

Some possible Job roles:

- Data Administrator
- Data Analyst
- Data Scientist
- Data Engineer
- Machine Learning Engineer
- Administrative
- Domain expert

…

e.g. European ICT Professional role profiles
https://tinyurl.com/ICT-profiles
https://tinyurl.com/ICT-salary
https://tinyurl.com/e-CF-methodology
https://tinyurl.com/e-CF-explore

If the role is not available, see other websites e.g.

- | Esco (europa.eu)
- Machine Learning Engineer Salary in Italy in 2023 | PayScale
- The Complete Machine Learning Engineer Salary Guide for 2023 (careerfoundry.com)
- Salary: Machine Learning Data Analyst in United States 2023 | Glassdoor
- Machine Learning Engineer Salaries 2023: A Comprehensive Guide | DataCamp
- Machine Learning Analyst Salary | Salary.com
- Machine Learning Engineer Salary-The Ultimate Guide for 2023 (projectpro.io)
- Machine Learning Engineer Salary: How Much Can You Make? | Coursera

Take all salaries and divide by the minimum (which will be 1) Example of detailed use case:

USER is a Data Analyst

STEP	COST CALCULATION	SC
1 USER open data balancing form	1 × 1 × 1.39	1.39
2 SYSTEM show the report		
3 USER check balancing condition	1 × 2 × 1.39	2.78
3.1 IF balancing is true	.9	
3.1.1 USER select YES	.9 × 1 × 1.39	1.25
3.2 ELSE	.1	
3.2.1 USER select NO	.1 × 1 × 1.39	0.14
4 USER select SUBMIT	1 × 1 × 1.39	1.39
5 SYSTEM show a confirmation		
6 USER close data balancing form	1 × 1 × 1.39	1.39
	NON-AUTOMATIO COST	**8.34**

STEP COST = OCCURENCE × COGNITIVE × SALARY

Where:

SALARY is the normalized job salary

COGNITIVE is the cognitive effort (see the file “From performance to productivity evaluation”)

OCCURRENCE is the no. of executions of the step. This number can be:

a) constant: e.g. “FOR EACH class…”, the number of classes is prefixed;

b) parametric: e.g., “ongoing validation” yes/no, it depends on the number of hyperparameters combinations.

c) empirical, e.g., “classes balanced” yes/no, it depends on the number of sessions to collect (c) versus the number of sessions needed (n), on the balancing tolerance, and so on. For example, for c>>n, which is very costly, the chance of having balanced classes is 100%, whereas for c < n it is 0%. Make the following assumptions:

- *balanced classes*, and *no. of iterations correct*, in 5 iterations, then 4 occurrences ko and 1 ok, i.e. 1/5 = 20% chance ok.
- *input coverage* in 3 iterations, 2 occurrences ko and 1 ok, then 1/3 = 33% ok.
- validation result 95% fine
- test result 99% fine

- *evaluation fine* for 6 iterations, 6 occurrences ok and 1 ko, the 1/7 = 14% ok

Assume 500 sessions for development, 5000 sessions for production, 50 sessions under evaluation.

Horizon: complete development, production and evaluation of 5 classifiers.